

3.1 Cryptography

Introduction to Cryptography and Cryptanalysis

Cryptography

Cryptography is the science of securing information by converting plaintext into ciphertext so that unauthorized users cannot understand it.

Basic Terms

Term, Meaning

Plaintext, Original readable message

Ciphertext, Encrypted message

Encryption, Converting plaintext into ciphertext

Decryption, Converting ciphertext back to plaintext

Key, Secret value used in encryption and decryption

Cipher, Algorithm used for encryption

Cryptanalysis

Cryptanalysis is the process of analyzing and breaking cryptographic systems to recover plaintext or keys without authorization.

Objectives of Cryptography

Confidentiality

Integrity

Authentication

Non-repudiation

2. Cryptanalysis

Cryptanalysis is the study and practice of breaking cryptographic systems without knowing the secret key.

Definition

Cryptanalysis is the process of analyzing encrypted information to recover the plaintext or secret key without authorization.

A person who performs cryptanalysis is called a **cryptanalyst**.

Objectives of Cryptanalysis

- Recover the original message.
- Discover the secret key.
- Identify weaknesses in encryption algorithms.
- Evaluate the security strength of cryptographic systems.

Example

Suppose an attacker intercepts:

KH00R

The attacker tries different shifts and discovers that shifting back by 3 gives:

HELLO

This process is cryptanalysis.

3.2 Symmetric Key Cryptography

Symmetric encryption uses the same key for encryption and decryption.

Process

Plaintext → Encryption Key → Ciphertext

Ciphertext → Same Key → Plaintext

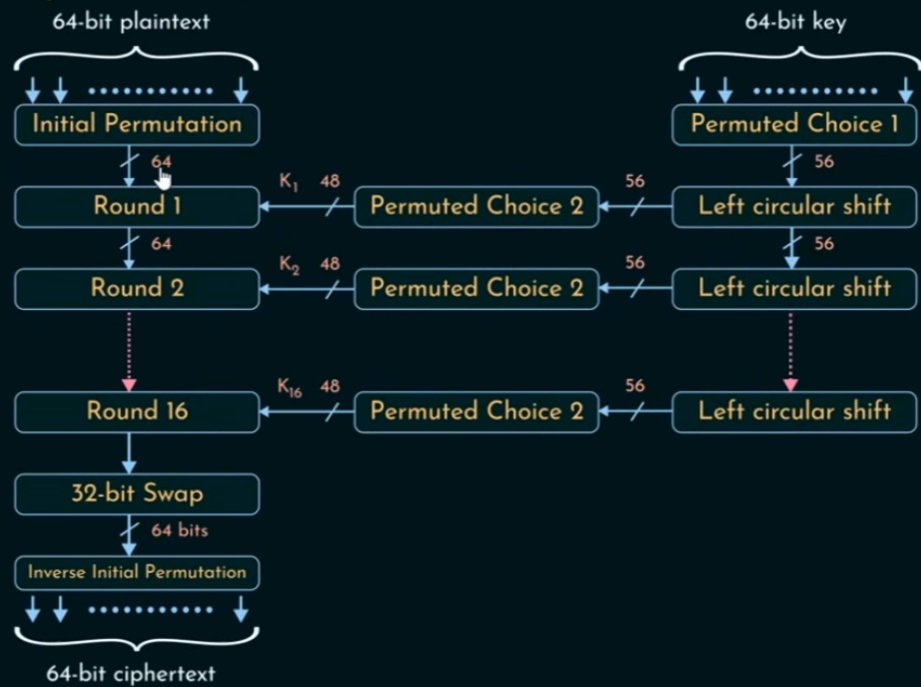
Advantages

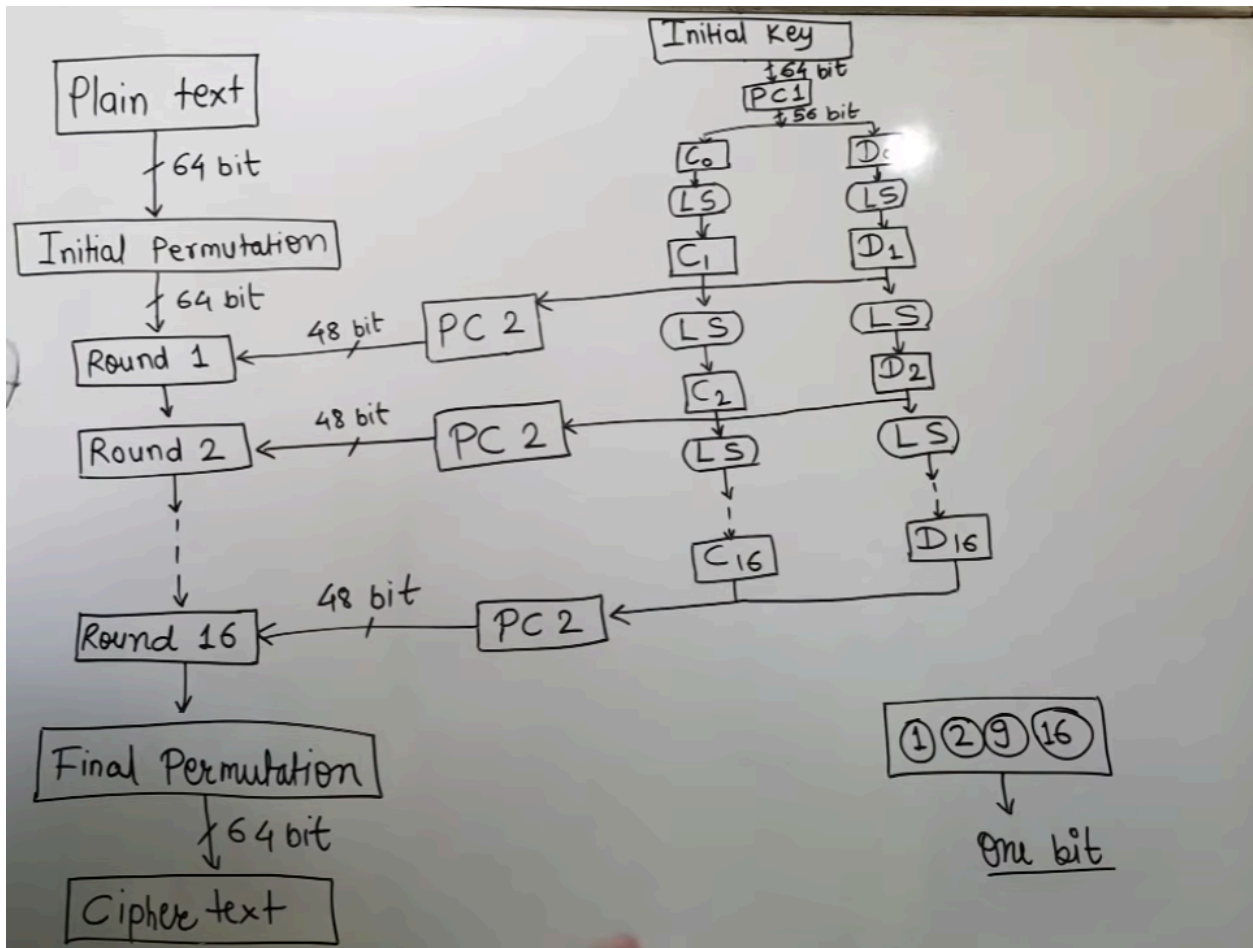
- Fast execution
- Suitable for large data

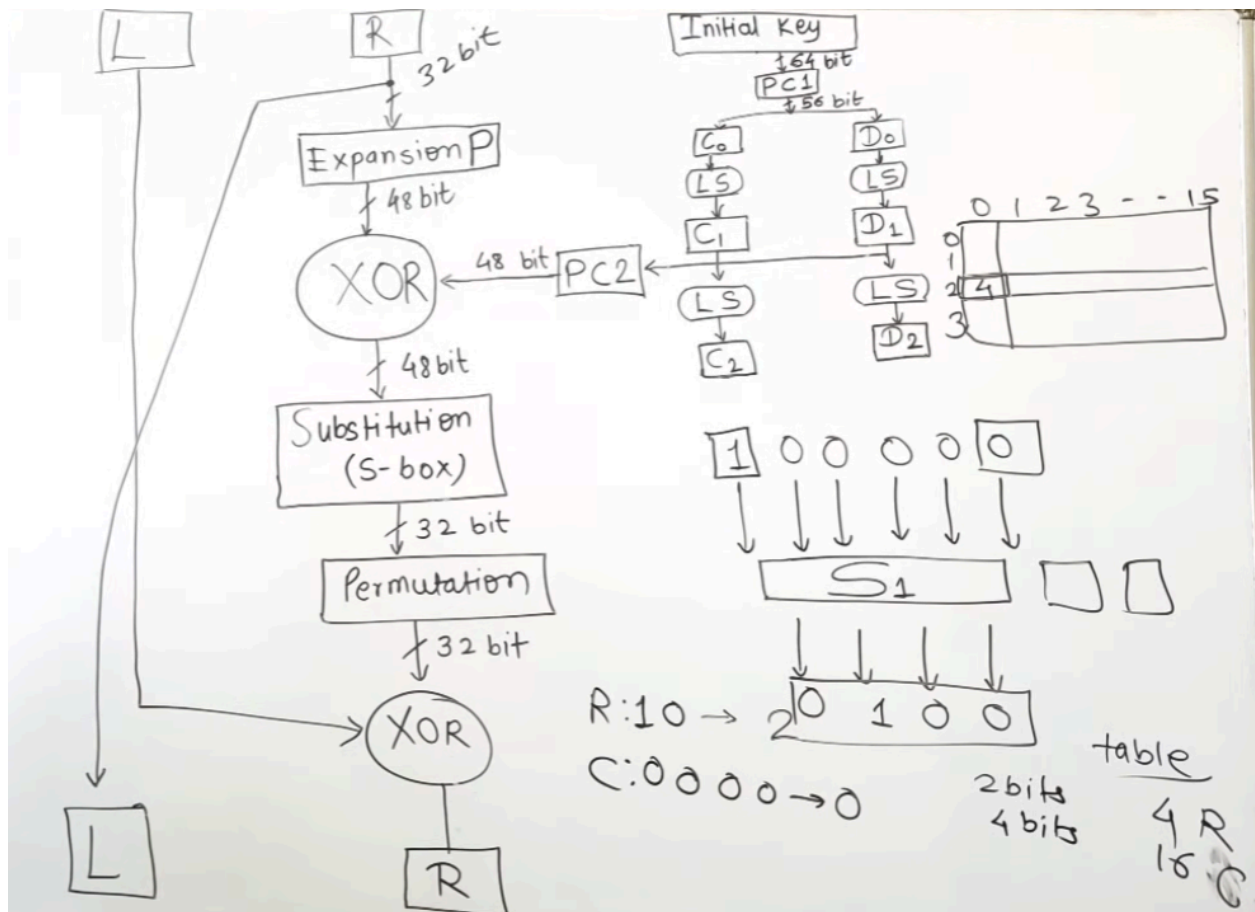
Disadvantages

- Key distribution problem
- Less scalable

DES Encryption Algorithm







Data Encryption Standard (DES Encryption)

- First developed in 1974 by IBM and NSA (based on LUCIFER that was 64-bit encryption with 128 bit key size).
- Also known as Data Encryption Algorithm (DEA) that was adopted in 1977 by National Bureau of Standards (presently called NIST) as the Data Encryption Standard.
- Data are encrypted in 64-bit blocks using a 56-bit key.
- The same steps, with the same key, are used to reverse the encryption.
- DES has the same structure as Feistel cipher except the F function and additional initial and final permutations IP and IP⁻¹, respectively.

Step1: Input plain text

64 bit plain text as input

Step2: Initial Permutation

Only bit pattern changed.

Step3: Divide into two halves

$L_0 \rightarrow 32$ bits

$R_0 \rightarrow 32$ bits

Step4: Key Preparation

- 64 bit Key
- Permitted Choice 1(PC1)- removes 8 parity bits $\rightarrow$ 56 bits
- Split into two - 28 bits halves
- Prefer left Circular Shift
- Apply Permitted Choice2 (PC2) to generate 48bits round key
- This is repeated to generate 48bits round key.

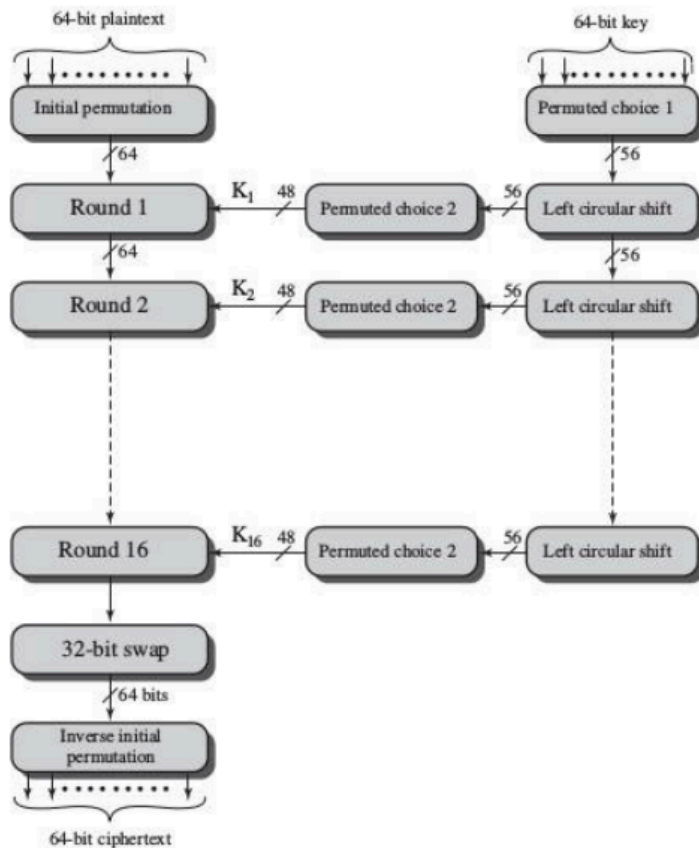
Step5: Round of Encryption

- Expand R_{i-1} (32bits $\rightarrow$ 48bits) using expansion Permutation
- XOR the expanded R with Round key k_i
- Pass the result through S-Boxes (48bits - 32bits)
- Apply permutation (P-Boxes)
- XOR with L_i to produce new R_i
- Set: $L_i = R_{i-1}$

R_i = result of XOR

Step6: 32bits swap

Step7: Inverse Initial Permutation



AES Overview

Unlike Feistel cipher, AES encrypts all 128 bits in one iteration. AES consists of three different types of layers, each of them manipulates all 128 bits of the data path (also called states).

(Each round, with the exception of the first, consists of all three layers.)

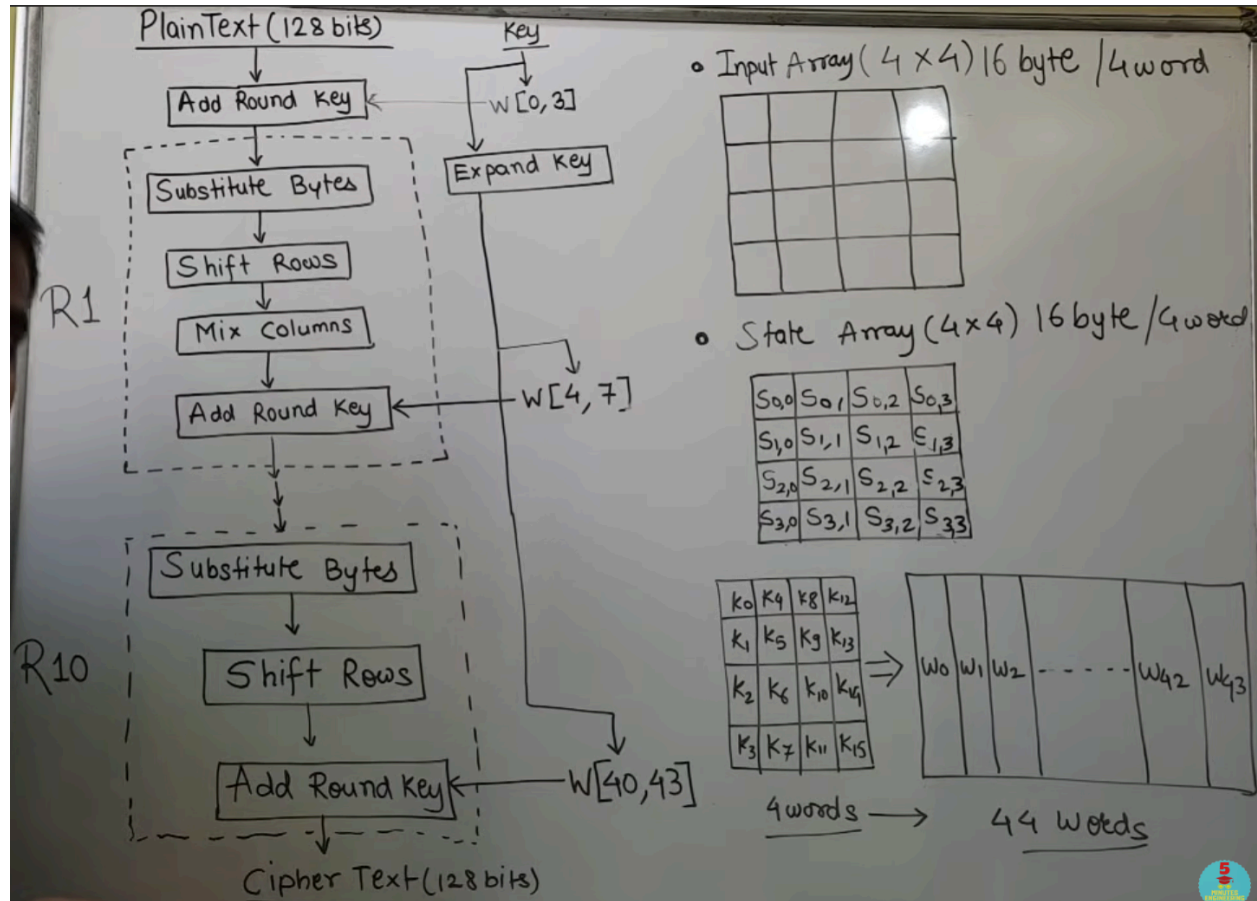
Key Addition layer A 128-bit round key, or subkey, which has been derived from the main key in the key schedule, is XORed to the state (or data path).

Byte Substitution layer (S-Box) Each element of the state is nonlinearly transformed (by providing confusion) using lookup tables with special mathematical properties.

Diffusion layer It provides diffusion over all state bits. It further consists of two sublayers:

- The ShiftRows layer permutes the data on a byte level.
- The MixColumn layer combines blocks of four bytes.

Key Scheduling Similar to DES, the key schedule computes round keys, or subkeys, $(k_0, k_1, \dots, k_{nr})$ from the original key.



Diffie-Hellman

Steps for Diffie-Hellman Key Exchange

1. Public Parameter Agreement

Alice and Bob publicly agree on a large prime p and a generator g ($1 < g < p$).

Everyone (including Eve) knows p and g .

2. Private Key Selection

Alice secretly chooses a random private integer a (her private exponent), where $1 < a < p$.

Bob secretly chooses a random private integer b (his private exponent), where $1 < b < p$.

3. Public Key Computation and Exchange

Alice computes her public value: $A = g^a \bmod p$ and sends A to Bob.

Bob computes his public value: $B = g^b \bmod p$ and sends B to Alice.

(Eve sees A and B , along with p and g .)

4. Shared Secret Computation

Alice receives B and computes the shared secret: $K = B^a \bmod p$

Bob receives A and computes the shared secret: $K = A^b \bmod p$

Why does this work?

Because of the mathematical property of exponents:

$$(g^b)^a \bmod p = g^{(b*a)} \bmod p$$

SHA Family		
Algorithm	Output Size	Developed by National Security Agency
		Types
SHA-1	160 bits	Current Recommendation SHA-256 and SHA-512 are widely used.
SHA-224	224 bits	
SHA-256	256 bits	
SHA-384	384 bits	
SHA-512	512 bits	

Digital Signatures

A digital signature is a cryptographic technique used to verify the authenticity, integrity, and non-repudiation of a digital message or document. It ensures that the message was created by a known sender and that it has not been altered during transmission.

Digital signatures verify:

- Sender authenticity
- Data integrity
- Non-repudiation

Working

1. Message is hashed.
2. Hash is encrypted using sender's private key.
3. Receiver verifies using sender's public key.

Benefits

- Authentication
- Integrity

- Non-repudiation

Digital Certificates

A digital certificate is an electronic document that verifies ownership of a public key.

Contains

- Owner's name
- Public key
- Certificate Authority signature
- Validity period

Example

HTTPS websites use digital certificates.

Public Key Infrastructure (PKI)

PKI is a framework that manages:

- Public keys
- Certificates
- Authentication

Components

1. Certificate Authority (CA)
2. Registration Authority (RA)
3. Digital Certificates
4. Public/Private Keys

Functions

- Secure communication
- User authentication
- Certificate management

Secure Communication Protocols

SSL (Secure Sockets Layer)

SSL was the original protocol used to secure communications over networks.

Functions

- Encryption
- Authentication
- Data integrity

Status

Largely replaced by TLS.

TLS (Transport Layer Security)

TLS is the modern replacement for SSL.

Features

- Strong encryption
- Secure key exchange
- Authentication

Uses

- Online banking
 - Email security
 - Web security
-

HTTPS

HTTPS stands for:

HyperText Transfer Protocol Secure

Formula

HTTPS=HTTP+TLS
HTTPS = HTTP + TLS
HTTPS=HTTP+TLS

Benefits

- Encrypts web traffic
- Prevents eavesdropping
- Protects user credentials

Example

When visiting a secure website, the browser displays a padlock icon.

- SSL/TSL provides confidentiality using symmetric encryption, data integrity and message authentication between server and client.

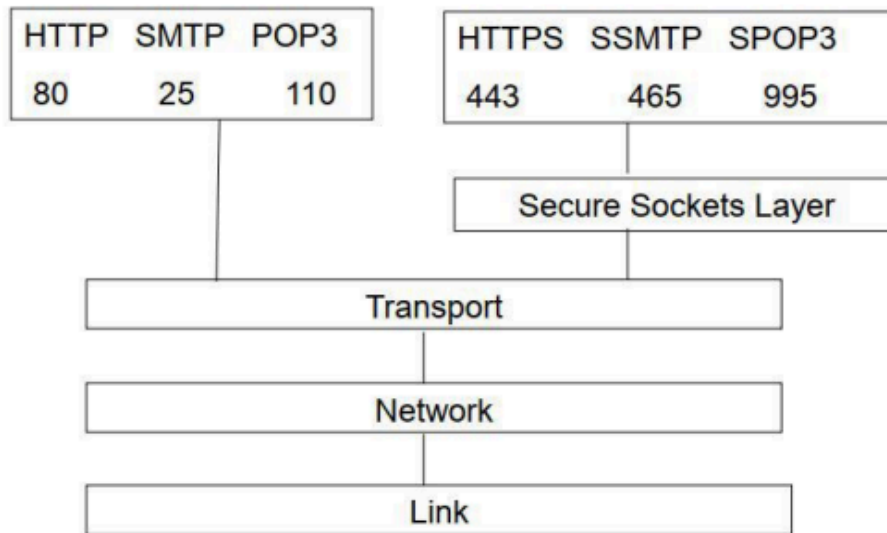


Figure: SSL uses.

SSL Architecture

- SSL has two layers of protocols which are implemented above the TCP layer.
- The SSL Record Protocol provides basic security services to various higher layer protocols e.g. HTTP.
- Three higher-layer protocols are defined as part of SSL:
 - i. Handshake protocol
 - ii. Change cipher Spec Protocol
 - iii. Alert Protocol

Two important concepts of SSL:

i. SSL connection:

- A transient peer-to-peer communication link that provides a suitable type of service.
- Every connection is associated with one session.

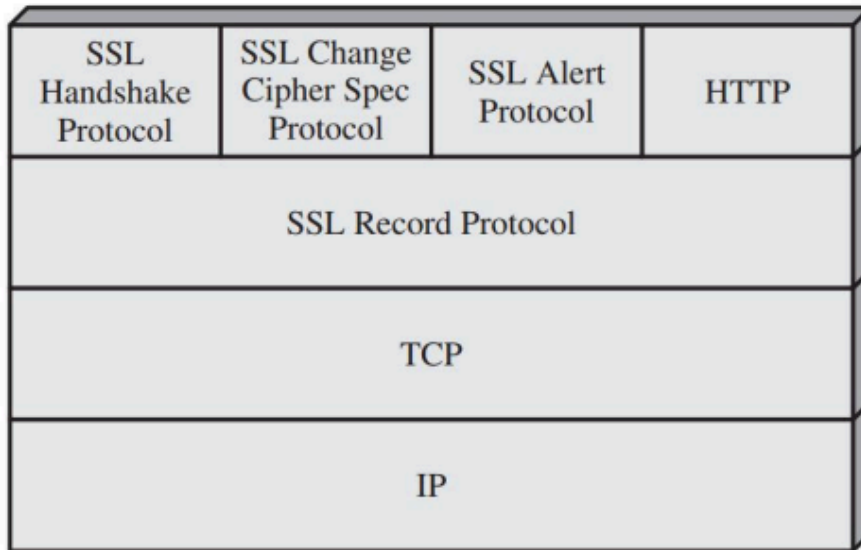


Figure: SSL protocol stack

ii. SSL session:

- An SSL session is an association between a client and a server.
- Created by handshake protocol.

SSL Record Protocol

- SSL Record Protocol provides two services for SSL connections:
 - Confidentiality:** The Handshake Protocol defines a shared secret key that is used for conventional encryption of SSL payloads.
 - Message Integrity:** The Handshake Protocol also defines a shared secret key that is used to form a message authentication code (MAC).

SSL Change Cipher Spec Protocol:

- One of 3 SSL specific protocols which use the SSL Record protocol.
- A single message with the value of 1.
- Causes pending state to become current, hence updating the cipher suit in use.

SSL Alert Protocol:

- Used to convey SSL-related alerts to the peer entity.
- SSL alerts are compressed and encrypted like all SSL data.

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Cyber Security

Unit 3 Cyber Threats and Attack Techniques

- Each message in this protocol consists of two bytes. The first byte takes the value warning (1) and fatal (2) to convey the severity of the message. If the level is fatal, SSL immediately terminates the connection.
- **Specific alert:** Unexpected message, decompression failure, handshake failure, illegal parameters.

Transport Layer Security (TLS)

Transport Layer Security (TLS) is a cryptographic protocol designed to provide security at the transport layer. It was derived from a security protocol called Secure Socket Layer (SSL). TLS ensures that no third party can eavesdrop on or tamper with any message transmitted between a client and a server.

Benefits of TLS

1. **Encryption:** TLS/SSL helps secure transmitted data using encryption, ensuring that sensitive information remains confidential and protected from unauthorized access.
2. **Interoperability:** TLS/SSL works seamlessly with most web browsers, including Microsoft Internet Explorer, and operates across a wide range of operating systems and web servers.
3. **Algorithm Flexibility:** TLS/SSL provides flexibility in choosing authentication mechanisms, encryption algorithms, and hashing algorithms used during secure sessions, allowing systems to adapt to different security requirements.
4. **Ease of Deployment:** Many applications can temporarily implement TLS/SSL on Windows Server 2003 operating systems, making it relatively easy to deploy in various environments.
5. **Ease of Use:** Since TLS/SSL operates beneath the application layer, most of its processes are completely invisible to the client, ensuring a smooth and user-friendly experience.

Working of TLS

When a client connects to a server using TCP, it initiates a process called the SSL/TLS handshake. During this handshake, the client sends several pieces of information to the server, including:

- The version of SSL/TLS it supports.
- A list of cipher suites (encryption algorithms) it can use.
- The compression methods it supports.

This allows the server to select compatible settings and establish a secure connection.

The server checks the highest SSL/TLS version supported by both parties, selects a cipher suite from the client's options (if compatible), and optionally chooses a compression method. After this setup, the server provides its digital certificate, which must be trusted either directly by the client or by a trusted third party.

HTTP vs HTTPS

Feature	HTTP	HTTPS
Security	Data is sent as plain text — easy to intercept.	Data is encrypted — safer from hackers.
Port	Uses port 80 by default.	Uses port 443 by default.
Trust	Browsers may label it “Not Secure.”	Shows a padlock in the address bar.
Speed	Slightly slower on old systems.	Often faster due to newer protocols and optimizations.
Use case	Suitable for public or non-sensitive info.	Best for sensitive data like payments, logins, personal info.

Why Choose HTTPS over HTTP?

1. Security

HTTPS is like sending your message in a locked box that only the receiver can open. This keeps sensitive information, like passwords or credit card numbers, safe from hackers.

2. Trust & Authority

Web browsers show a padlock icon for HTTPS websites. People trust these sites more, and search engines also rank them higher. So, HTTPS can help you win both visitors' trust and better search results.

3. Speed & Better Tracking

HTTPS websites load faster than HTTP also helps website owners see where their visitors come from (like social media or ads) more accurately, which is useful for improving their content.

IPSec

- IPSec stands for Internet Protocol Security.
- It is a collection of protocols to provide security for a packet at the network layer.
- IPSec is a standard framework for ensuring private communication over public network.
- It has become the most common network layer security control, typically used to create a VPN.

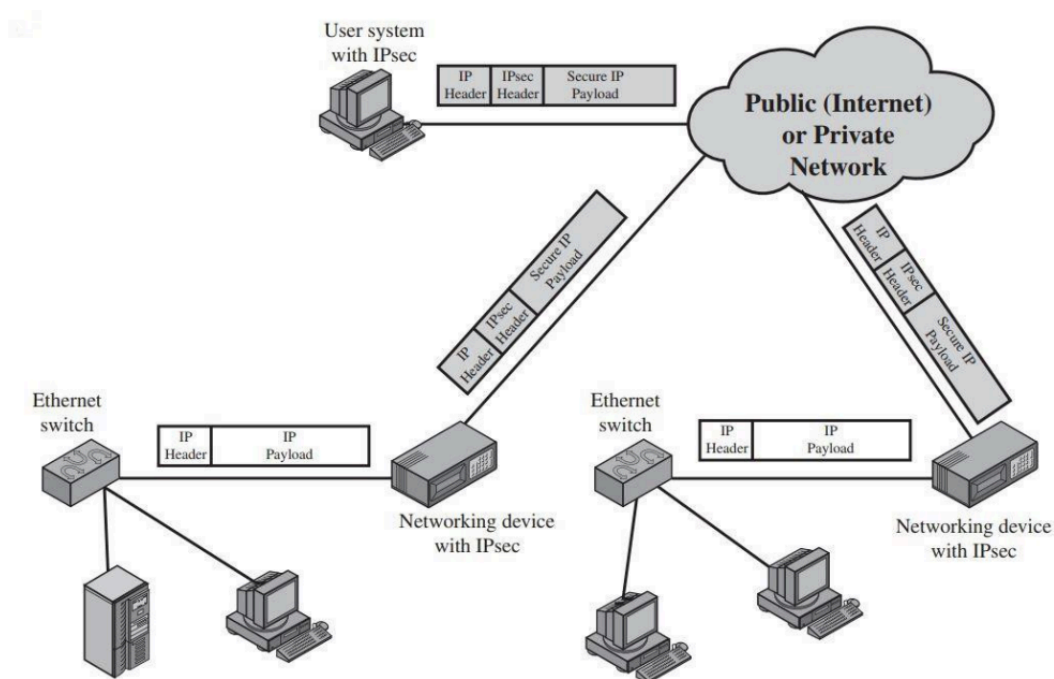


Figure: An IPSec scenario.

Modes of IPSec

IPSec operates in one of two different mode: transport mode and tunnel mode. Transport Mode

- Transport mode protects the payload to be encapsulated in the network layer i.e. it protects what is delivered from the transport layer to the network layer.

- It does not protect the whole IP header or in other words, it does not protect the whole IP packet.

Tunnel Model

- In tunnel mode, IPSec protects the entire IP packet. It takes an IP packet, including the header, applies IPSec security methods to the entire packet, and then adds a new IP header

8.16.1.2 IPSec Components

- Two security protocols :

- Authentication Header (AH): AH is an extension header to provide message authentication.

Because message authentication is provided by ESP, the use of AH is deprecated.

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Cyber Security

Unit 3 Cyber Threats and Attack Techniques

- Encapsulating Security Payload (ESP): ESP consists of an encapsulating header and trailer used to provide encryption or combined encryption/authentication and integrity protection

- Internet Key Exchange (IKE) protocol

- IPSec uses IKE to negotiate IPSec connection settings.

- Authenticate endpoints to each other.

- Define the security parameters of IPSec-protected connections

- Negotiate secret keys and

Firewall

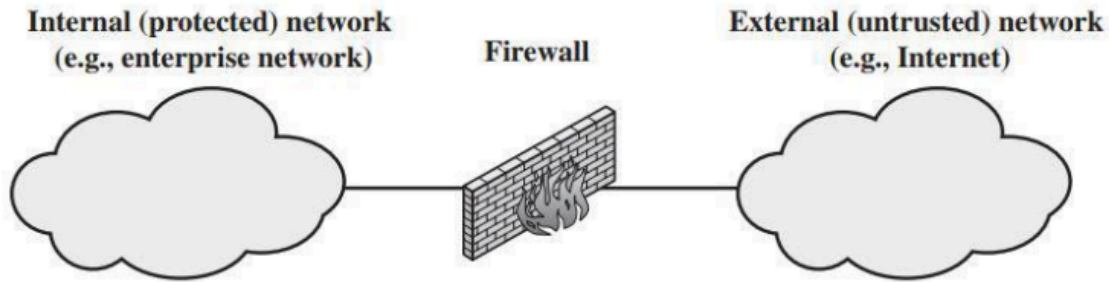


Figure 8.31: A general model of firewall.

- Firewall is a device (usually a router or computers) installed between the internal network of an organization and the rest of the Internet.
- A firewall is a network security system designed to prevent unauthorized access to or from a private network.
- A firewall security policy dictates which traffic is authorized to pass in each direction.
- A firewall may be designed to operate as a filter at the level of IP packets, or may operate at a higher protocol layer.
- A firewall provides a location for monitoring security-related events as a single-choke point.
- Auditing and controlling access can implement alarms for abnormal behavior.

Firewalls

A firewall is a security device or software that monitors and filters incoming and outgoing network traffic.

Functions

- Blocks unauthorized access
- Filters network traffic
- Protects internal networks

Types of Firewalls

1. Packet Filtering Firewall

- Examines packets individually
- Uses IP addresses and ports

2. Stateful Inspection Firewall

- Tracks active connections
- More secure than packet filtering

3. Proxy Firewall

- Acts as intermediary between users and servers

4. Next Generation Firewall (NGFW)

- Deep packet inspection
- Intrusion prevention
- Application awareness

Firewall Configuration

Common Rules

- Allow trusted traffic
- Block suspicious traffic
- Restrict unused ports
- Monitor logs regularly

Virtual Private Networks (VPNs)

VPN creates a secure encrypted tunnel over a public network.

Benefits

- Privacy protection
- Secure remote access
- Data encryption
- Safe internet browsing

Working

User → Encrypted Tunnel → VPN Server → Internet

Types of VPN

Remote Access VPN

Allows individual users to connect securely to an organization.

Site-to-Site VPN

Connects two or more networks securely.

Applications

- Remote work

- Secure business communication
- Accessing private corporate resources

Exam-Oriented Short Notes

Cryptography

Technique of securing information through encryption.

Cryptanalysis

Study of breaking cryptographic systems.

AES

Modern symmetric encryption standard using 128, 192, or 256-bit keys.

RSA

Popular asymmetric algorithm used for encryption and digital signatures.

Hash Function

One-way function producing fixed-size output.

Digital Signature

Mechanism for authentication, integrity, and non-repudiation.

PKI

Framework for managing public keys and digital certificates.

TLS

Modern protocol for securing internet communication.

IPSec

Protocol suite securing IP communications.

Firewall

Security device that filters network traffic.

VPN

Secure encrypted connection over a public network.